

Analog Lab Report Week 2

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Objective :

This experiment aims to design, derive, and implement a single-ended to differential input converter alongside a PWM modulator. The circuit will be modelled and simulated in LTspice to verify the mathematical derivations, followed by physical hardware implementation to validate the simulated performance.

List of Components :

1. For the ramp signal we will use the previous circuit.
2. 1 - LM339
3. 1 – MCP6004
4. 2 – 10k ohm Pull up resistors
5. 2 – 10k ohm resistors
6. 1 – 10 micro Farad Capacitor
7. 2 – 10n Farad Capacitors for low pass filtering
8. 2 – 100k ohm Resistors for low pass filtering
9. Power Supply (ADALM 1000)
10. Breadboard

EXPERIMENT-2: SINGLE ENDED-TO-DIFFERENTIAL INPUT CONVERTER AND PWM MODULATOR

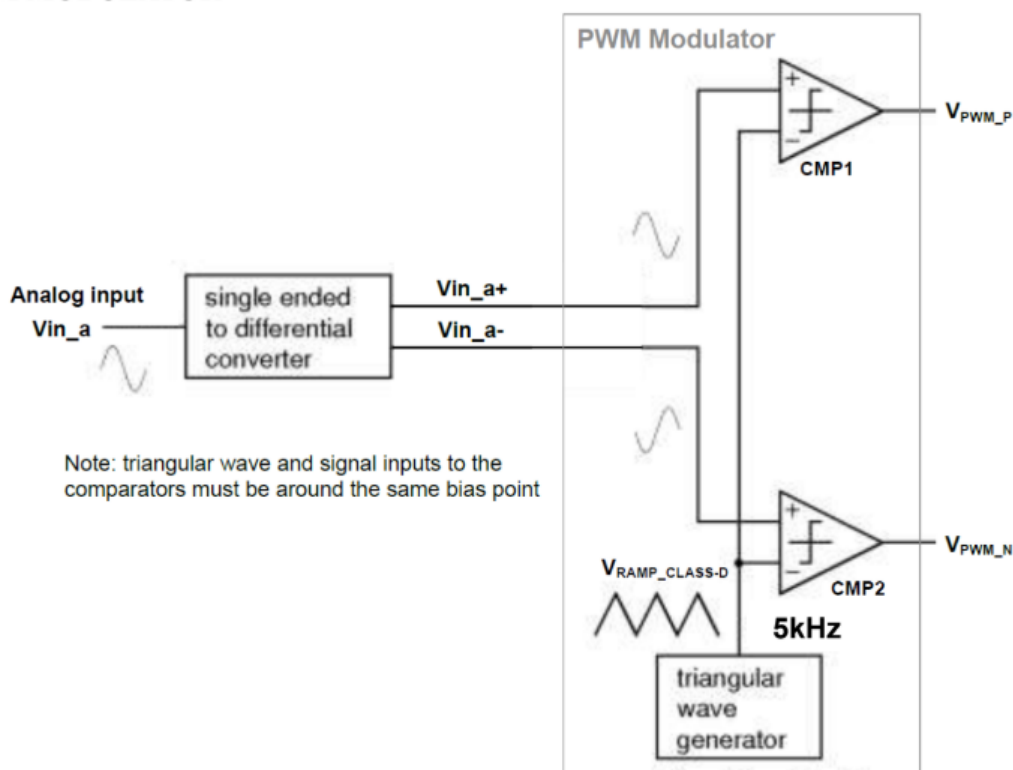


Figure 2 Schematic diagram of single-to-differential input converter and PWM modulator.

Single ended-to-differential converter can be designed using op-amp based inverting amplifier as shown in Figure 2.

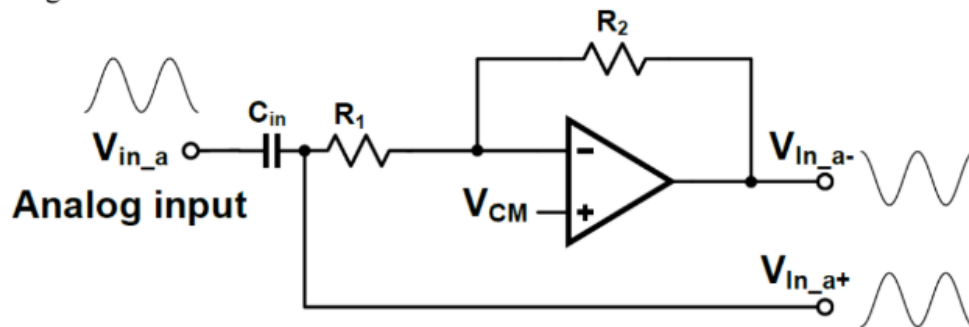


Figure 3 Schematic diagram of single-to-differential converter.

Derivation of Formula :

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Here, we are considering a high value capacitor so that V_{in-a} passes through without alteration or distortion.

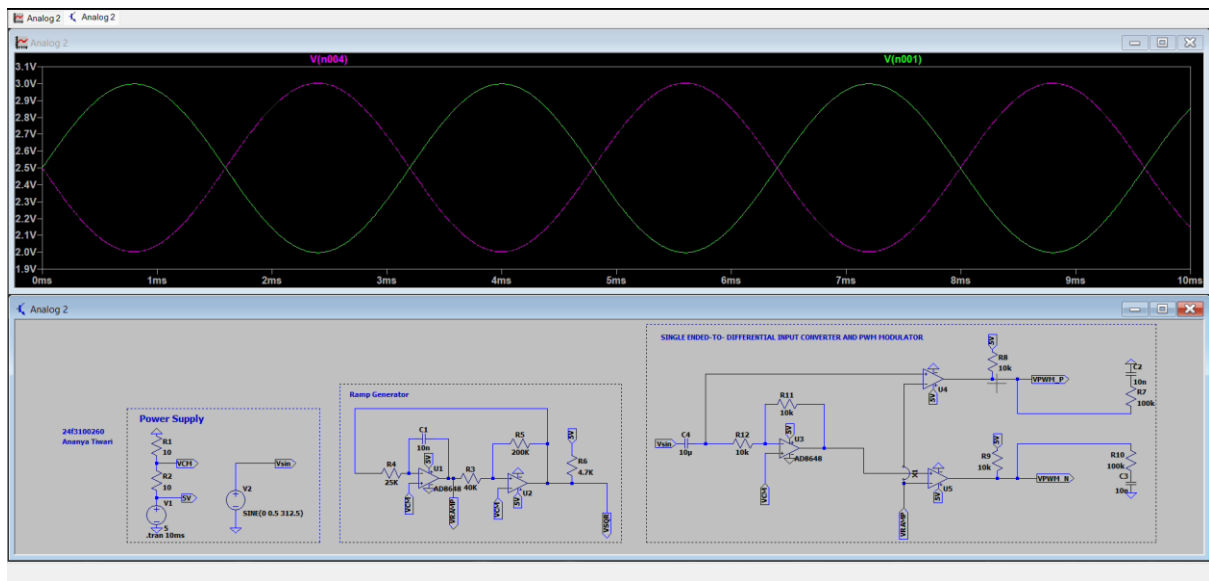
$C_{in} = 10 \mu f$ for our experiment

$$V_{in-a-} = -\frac{R_2}{R_1} \times V_{in-a}$$

$$V_{in-a+} = V_{in-a}$$

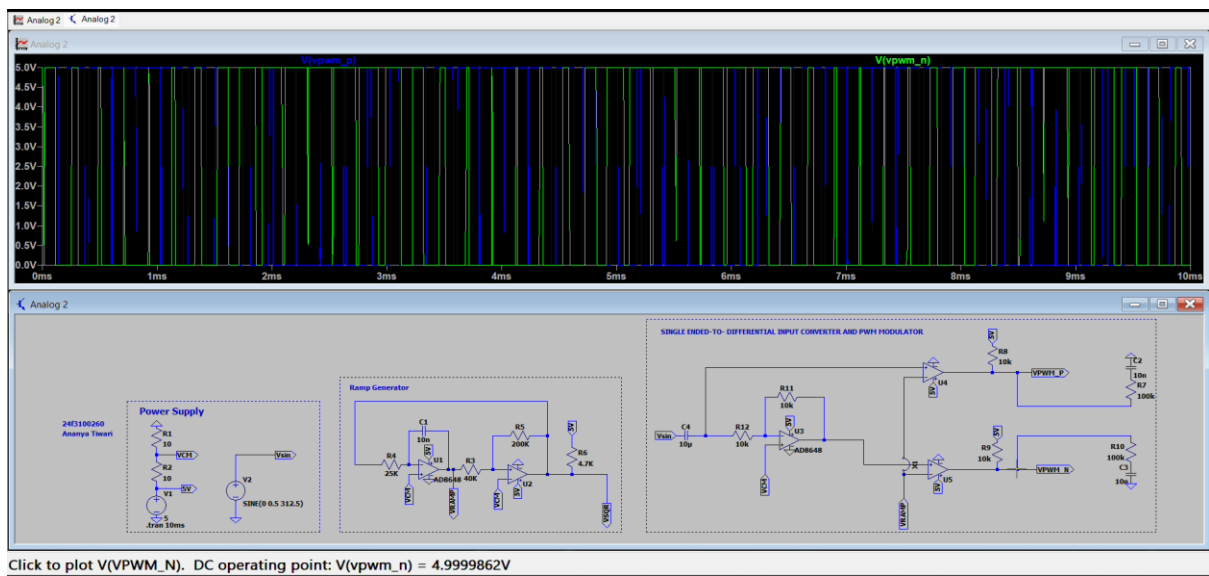
V_{in-a+} & V_{in-a-} have same amplitude by this equation but in reality there will be more factors which will affect.

LTspice Schematics and Output :

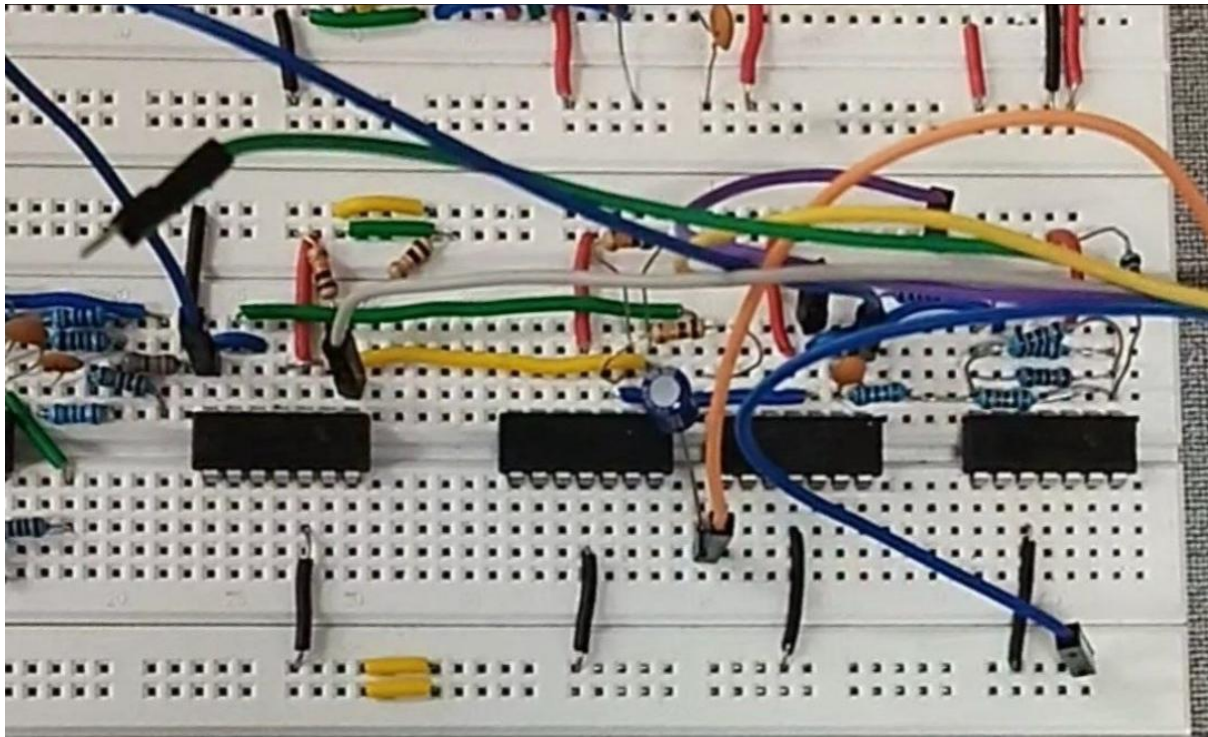


Green wave - V_{in_a+}

Pink wave – V_{in_a-}



Physical Circuit :



Physical Circuit Wire Connections :-

Green wire and Blue wire – GND

Yellow – 5V

Orange wire – Channel B (*input*)

White wire – Channel A (*output*)

Purple wire – 2.5V

Physical Circuit Output :-



First one – V_{PWM_P}

Second one – $V_{\{PWM_N\}}$



Conclusion :-

The experiment was successful, with the physical hardware output validating the simulation models.